

CSA 0378-DATA STRUCTURES

14. LINEAR AND BINARY SEARCH

```
main.c
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, key, i, found = 0;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter element to search: ");
15    scanf("%d", &key);
16
17    for(i = 0; i < n; i++)
18    {
19        if(a[i] == key)
20        {
21            found = 1;
22            printf("Element found at position %d\n", i + 1);
23            break;
24        }
25    }
26
27    if(found == 0)
28        printf("Element not found\n");
29
30    return 0;
31 }
```

Output

```
Enter number of elements: 4
Enter elements:
10
20
30
40
Enter element to search: 20
Element found at position 2

=== Code Execution Successful ===
```

```
main.c
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[100], n, key;
6     int low, high, mid;
7     int found = 0, i;
8
9     printf("Enter number of elements: ");
10    scanf("%d", &n);
11
12    printf("Enter sorted elements:\n");
13    for(i = 0; i < n; i++)
14        scanf("%d", &a[i]);
15
16    printf("Enter element to search: ");
17    scanf("%d", &key);
18
19    low = 0;
20    high = n - 1;
21
22    while(low <= high)
23    {
24        mid = (low + high) / 2;
25
26        if(a[mid] == key)
27        {
28            found = 1;
29            printf("Element found at position %d\n", mid + 1);
30            break;
31        }
32        else if(key < a[mid])
33        {
34            high = mid - 1;
35        }
36        else
37        {
38            low = mid + 1;
39        }
40    }
41
42    if(found == 0)
43        printf("Element not found\n");
44
45    return 0;
46 }
```

Output

```
Enter number of elements: 4
Enter sorted elements:
10
20
30
40
Enter element to search: 20
Element found at position 2

=== Code Execution Successful ===
```

```
main.c
11 printf("Enter sorted elements:\n");
12 for(i = 0; i < n; i++)
13     scanf("%d", &a[i]);
14
15 printf("Enter element to search: ");
16 scanf("%d", &key);
17
18 low = 0;
19 high = n - 1;
20
21 while(low <= high)
22 {
23     mid = (low + high) / 2;
24
25     if(a[mid] == key)
26     {
27         found = 1;
28         printf("Element found at position %d\n", mid + 1);
29         break;
30     }
31     else if(key < a[mid])
32     {
33         high = mid - 1;
34     }
35     else
36     {
37         low = mid + 1;
38     }
39 }
40
41 if(found == 0)
42     printf("Element not found\n");
43
44 return 0;
45 }
```

Output

```
Enter number of elements: 4
Enter sorted elements:
10
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